

Project plan

LLM Hosting

Nahayat

Oisterwijk, Netherlands

Version

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Communication

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1. Project Assignment

1.1 Context

Nahayat is an AI consultancy in Oisterwijk, founded by Robin Damen, focused on practical AI for SMBs. Besides giving advice, we build and run the agents ourselves, using open-source components so clients keep control over cost and stack, with no vendor lock-in. We host everything through our sister company Virtual Computing's Dutch datacenters which matters especially with regards to the AVG/GDPR and EU AI act.

The company has developed and self-hosts its own AI orchestration layer, including agents, MCP servers and workflows. However, the Large Language Models (LLMs) used by these systems are primarily accessed through external cloud APIs, such as Azure OpenAI and AWS Bedrock/Anthropic. Although agreements and data-retention policies are in place to allow these services to be used safely, the company sees an opportunity to gain more control over its AI infrastructure by introducing self-hosted LLM inference.

1.2 Goal of the project

The goal of this project is to investigate to what extent cloud-based LLM APIs can be replaced or complemented by self-hosted LLMs within Nahayat's existing AI ecosystem. The investigation will initially cover all four use-case categories identified as relevant by the company:

- **Conversational/chat:** evaluating the ability of local LLMs to provide useful and coherent conversational responses.
- **Retrieval-Augmented Generation (RAG):** evaluating the ability of local LLMs to answer questions using retrieved information from an external knowledge source.
- **Classification and smaller LLM tasks:** evaluating local models on structured tasks where smaller and less resource-intensive models may be sufficient.
- **Agentic tool use / MCP:** evaluating whether local LLMs can reliably select and use external tools or functions within agentic workflows.

Selected self-hosted models will be tested across these use cases and compared, where applicable, with the cloud-based models currently used by Nahayat. The evaluation will consider response quality, performance, resource usage, reliability, integration feasibility and hardware requirements.

All four use cases are initially included in the investigation. If testing demonstrates that particular use cases or model sizes are not realistically feasible on the available hardware, the scope may be refined in consultation with the company supervisor. Such a decision will be supported by the obtained technical results and documented as a finding of the project.

The desired outcome is a technically validated understanding of which workloads can be handled effectively by self-hosted LLMs, which are better suited to cloud-based models, and where a hybrid approach may be appropriate.

1.3 The assignment

The assignment is to investigate, test and evaluate the feasibility of using self-hosted Large Language Models (LLMs) as an alternative or complement to the cloud-based LLM APIs currently used within the company's AI ecosystem.

The project will identify relevant AI use cases, select and evaluate suitable self-hosted models, determine what can be achieved with the currently available hardware, and compare their capabilities and performance with the existing cloud-based solutions. The investigation should include practical testing within relevant existing systems, such as agents, RAG applications, MCP/tool-based integrations and automated workflows.

Based on the results, the project will provide a substantiated recommendation indicating which workloads are suitable for self-hosted inference, which are better suited to cloud-based models, where a hybrid approach is appropriate, and under which circumstances an investment in additional hardware would be justified.

1.4 Scope

The scope of this project is to investigate the technical viability of running Large Language Models (LLMs) locally on hardware provided by Nahayat. The investigation will initially cover four use-case categories: conversational/chat, RAG, classification and smaller LLM tasks, and agentic tool use through MCP or similar tool-calling mechanisms.

The project includes researching and selecting suitable open-source/local LLMs, deploying them on the available hardware, developing the required benchmark or proof-of-concept implementations, and evaluating their performance across the four use cases. Where applicable, results will be compared with relevant cloud-based models currently used by Nahayat.

The project does not aim to develop complete production-ready applications or replace Nahayat's existing AI infrastructure. Implementations will primarily serve research, benchmarking and proof-of-concept purposes.

The four use cases remain in scope unless technical evidence, such as hardware limitations or insufficient model capability, demonstrates that further investigation of a particular case is not feasible. Any reduction in scope will be discussed and agreed with the company supervisor and documented together with the technical justification.

The project includes:	The project does not include:
1 Researching and selecting relevant open-source/local LLMs, primarily from sources such as Hugging Face.	1 Developing a complete production-ready AI platform or replacing Nahayat's existing AI infrastructure.
2 Setting up and configuring selected LLMs on the hardware provided by Nahayat.	2 Developing complete production applications for every tested use case. Implementations created during the project will primarily serve benchmarking and proof-of-concept purposes.

1.5 Conditions

Hardware availability

Nahayat will provide access to the hardware on which the local LLMs will be tested. The final selection of models and model sizes will depend on the specifications and capabilities of this hardware.

Benchmark environment

Nahayat will provide access to the benchmark environment together with the allocated hardware. This environment will be used as the main basis for testing and comparing models.

Model selection

The project will primarily focus on models that can be executed locally on the available hardware. Open-source models, particularly models available through Hugging Face, are preferred, but other suitable models and technologies may also be considered when relevant.

1.6 Finished products

The project will deliver a set of research, technical and evaluation products. Together, these products provide the evidence required to determine the technical and financial feasibility of using self-hosted LLMs within Nahayat.

1. Current AI Environment Analysis

An analysis of the relevant parts of Nahayat's current AI environment. This includes the current use of cloud-based LLMs, relevant AI use cases, existing integrations, technical requirements and constraints. The analysis will also support the selection of representative use cases for further testing.

2. LLM and Technology Research

Research into suitable self-hosted LLMs and supporting inference technologies. Candidate models will be investigated based on factors such as capabilities, hardware requirements, model size, quantization options and suitability for the selected use cases.

3. Benchmark and Test Framework

A reproducible testing setup for evaluating selected self-hosted LLMs and comparing them with relevant cloud-based models. The framework will contain representative test cases, evaluation criteria, configurations and benchmark scripts.

4. Local LLM Prototypes / Proofs of Concept

Working prototypes demonstrating selected self-hosted LLMs within representative use cases. Depending on the selected case studies and available hardware, these may include chat, RAG, classification or agentic/tool-use scenarios.

5. Benchmark Results and Evaluation Report

A documented evaluation of the tested models. Results will include relevant measurements such as response quality, latency, throughput, hardware resource usage, context limitations and tool-calling reliability.

6. Hardware Requirements Analysis

An analysis of the capabilities and limitations of the available hardware based on the benchmark results. Where the existing hardware is insufficient, the report will identify the hardware requirements necessary to support relevant models and workloads.

7. Financial Feasibility Analysis

An analysis comparing the estimated costs of operating a viable self-hosted LLM solution with the continued use of commercial cloud-based LLM services. Where technically relevant, potential hardware investments will be evaluated against expected usage and operating costs.

8. Final Feasibility Study and Recommendation

The main final product of the project, combining the technical, benchmark, hardware and financial findings. It will provide a substantiated recommendation describing which evaluated workloads are suitable for self-hosted inference, which are better suited to cloud-based models, where a hybrid approach is appropriate, and under which conditions additional hardware investment could be justified.

1.6 Research questions

“To what extent can locally-hosted open-source LLMs, on limited hardware, be used for chat, retrieval, classification, and agentic tool-use, and when would hardware investment be justified?”

- Which locally hosted LLMs can run acceptably on the available hardware, and what are their hardware requirements?
- How does the output quality of locally hosted LLMs compare with the commercial cloud-based models currently used by Nahayat across the investigated use cases?
- What technical limitations do locally hosted LLMs have regarding factors such as inference speed, context length, resource usage and tool-calling reliability?
- To what extent are conversational/chat tasks, RAG, classification and smaller LLM tasks, and agentic tool use technically viable with locally hosted LLMs?
- How do the estimated costs of locally hosted LLMs compare with commercial cloud-based LLM services for the investigated use cases?
- If self-hosted LLMs are technically viable, what hardware investment would be required, and under what expected usage and cost conditions would this investment be financially justified?
- What privacy, ethical, sustainability and future-proofing considerations influence the suitability of locally hosted LLMs within Nahayat's AI environment?

2. Approach and Planning

2.1 Approach

The project will follow an iterative and research-driven approach. Since the suitability of self-hosted LLMs, available technologies and hardware limitations cannot be fully determined beforehand, the project requires continuous experimentation, evaluation and adjustment based on the obtained results.

The project will start with an orientation and problem-analysis phase. During this phase, the current AI environment, use cases, cloud-based LLM usage, available hardware, requirements and constraints will be investigated together with relevant company stakeholders. Based on these findings, the project scope and research questions will be refined.

After the orientation phase, the project will continue in short iterative cycles. Each iteration will focus on a specific part of the investigation, such as model selection, deployment, integration, benchmarking or evaluation. Findings from one iteration can influence the priorities and technical choices of subsequent iterations.

Progress and findings will be discussed regularly with the company supervisor. Where possible, working results and experiments will be demonstrated to relevant stakeholders to obtain feedback.

The final phase will focus on completing the technical evaluation, validating the results, formulating recommendations, completing the project documentation and portfolio, and transferring the project results to the company

2.1.1 Test approach

Testing will be performed throughout the project rather than only at the end. Because the project focuses on evaluating the feasibility of self-hosted LLM inference, testing will include both technical testing and comparative evaluation.

2.2 Research methods

The research methods used in this project are structured according to the Development Oriented Triangulation (DOT) framework. The framework helps select and combine appropriate research methods from different research strategies. Within this project, Library methods are used for literature and technology research, Field methods for gathering information from stakeholders and analysing the existing environment, Workshop methods for prototyping, Lab methods for benchmarking and testing, and Showroom methods for validating results and prototypes with stakeholders. Combining these methods allows the research questions to be investigated from multiple perspectives and supports evidence-based conclusions.

Different ICT research methods will be combined throughout the project. The exact methods may be adjusted as the research questions and project scope are refined.

Stakeholder interviews / requirements gathering

Interviews and discussions with the company supervisor and other relevant stakeholders will be used to understand the current AI architecture, existing cloud LLM usage, important use cases, technical constraints and expectations for self-hosted inference.

Literature and technology research

Desk research will be used to investigate available self-hosted LLMs, inference frameworks, hardware requirements, model capabilities and relevant developments in local LLM deployment. Technical documentation, research publications and other reliable technical sources will be used to substantiate technology and model selection.

Existing system analysis

The company's existing AI infrastructure will be analysed to understand how LLMs interact with agents, RAG systems, MCP servers and workflows. This will help determine the technical requirements that a self-hosted inference solution must satisfy.

Prototyping

Selected self-hosted models and inference technologies will be deployed and integrated into representative use cases. Prototypes will be used to determine whether proposed solutions work within the company's actual technical environment.

Benchmarking and comparative testing

Selected self-hosted models will be tested against representative company workloads and, where applicable, compared with the currently used cloud-based models. Measurements may include response quality, latency, throughput, resource usage and task-specific success criteria.

Expert/stakeholder evaluation

Results, prototypes and recommendations will be reviewed with the company supervisor at relevant validation moments. The company supervisor will validate their practical relevance to Nahayat and provide feedback on important technical or scope decisions. Additional company experts may be consulted when specialist knowledge of a particular system or use case is required. Detailed stakeholder roles and involvement are documented separately in the Stakeholder Analysis.

2.3 Learning Outcomes

Learning Outcome 1 – Professional Duties

The project will demonstrate professional IT activities through analysis, advice, design, realization and management/control. Evidence will include the current AI environment and stakeholder analysis, research and benchmark documentation, local LLM prototypes and integrations, validation results, hardware and financial analyses, and the final advice report.

Learning Outcome 2 – Situation Orientation

The project is carried out within Nahayat's existing AI environment and requires previously acquired IT knowledge to be applied in a new technical and organizational context. Evidence will include adaptation to available hardware and company systems, use of company-specific tools and workflows, Jira planning, technical experimentation and documented adjustments to project decisions based on new findings.

Learning Outcome 3 – Professional Standard

The project applies the DOT research framework to investigate the feasibility of self-hosted LLMs in a complex and uncertain technical environment. Technical, organizational, financial, privacy, ethical and future-oriented considerations will be incorporated into the research and final recommendation. Evidence will include the research report, stakeholder involvement, benchmark results, feasibility analyses and advice report.

Learning Outcome 4 – Personal Leadership

Personal leadership will be demonstrated by independently planning and executing the project, setting development goals, requesting and applying feedback, documenting progress and adapting the approach when technical or organizational challenges occur. Evidence will include Jira/GitHub activity, feedback sessions, personal goals and the final reflection document.

2.4 Time plan

Phase 1 – Orientation and problem analysis

Understand the existing AI infrastructure, cloud LLM usage, available hardware, relevant use cases, stakeholder requirements and project constraints. Refine the project scope and research questions.

Phase 2 – Technology and model research

Investigate suitable self-hosted LLMs, inference frameworks, deployment approaches and hardware requirements. Select promising candidates and representative company use cases for further evaluation.

Phase 3 – Prototype and integration

Deploy selected self-hosted models and develop the required integrations to test them within representative parts of the company's existing AI ecosystem.

Phase 4 – Benchmarking and evaluation

Evaluate the selected solutions using defined test cases and compare the results with

relevant cloud-based models. Analyse model quality, performance, hardware utilization, integration feasibility and other agreed evaluation criteria.

Phase 5 – Refinement and validation

Improve or adjust the prototypes and experiments based on previous findings. Validate important results and recommendations together with relevant company stakeholders.

Phase 6 – Conclusion and completion

Determine which workloads are suitable for self-hosted inference, which should remain cloud-based and where a hybrid approach is preferable. Finalize recommendations regarding hardware requirements, complete documentation and portfolio evidence, and transfer the project results to the company.

Phasing	Effort	Start	Ready
1 Orientation	15%	31-08-2026	13-09-2026
2 Research	15%	14-09-2026	11-11-2026
3 Prototype	25%	11-11-2026	22-11-2026
4 Benchmarking	20%	22-11-2026	13-12-2026
5 Validation	10%	13-12-2026	24-01-2027
6 Conclusion	15%	24-01-2027	31-01-2027

3. Project Organization

3.1 Team members

Name + Phone + e-mail	Role/tasks	Availability
Luc Tortike 0639881390 lmtortike@nahayat.io	Project Supervisor	Every other week, on Tuesday.
Thayyakathe Raveendran, Sreedevi S. 0885085697 s.thayyakatheraveendran@fontys.nl	Teacher	Tuesday and Wednesday 13:00 – 16:00
Marian-Metin Constantin 0687289257 metin.constantin@student.fontys.nl	Student	Mon, Tue, Thu, Friday 09:00 – 16:00

3.2 Communication

Communication between the student and company is realized through mail or physically at the company location.

Communication between the student and teacher is realized through Microsoft teams.

3.3 Test environment

Testing is performed on the hardware and benchmark environment provided by Nahayat. The environment is used to deploy and evaluate locally hosted LLMs under controlled conditions.

Different models and model sizes will be tested on the available hardware while monitoring performance indicators such as response speed, resource utilization and model behavior. The same environment will also be used for the development and testing of proofs of concept for the selected use cases, including chat, RAG, classification and agentic tool use.

The hardware configuration, model versions, quantization levels and relevant inference settings will be documented during testing to ensure that benchmark results can be reproduced and compared. Cloud-based LLMs currently used by Nahayat will be used as comparison baselines where applicable.

3.4 Configuration management

Project files and source code will be managed using GitHub as the central version-control repository. The repository will contain project documentation, benchmark scripts, model configurations, prototype implementations and other technical artifacts produced during the project.

Changes to source code and configurations will be tracked using Git so that experiments and implementations can be traced to specific versions. Model files and large model weights will not be stored directly in the repository. Instead, the model name, version, quantization and relevant configuration used during an experiment will be documented.

The Jira board will be used alongside GitHub to manage project activities, tasks and progress. Relevant benchmark configurations and results will be documented so that important experiments can be reproduced and linked to the corresponding project activities.

4. Finance and Risks

4.1 Cost budget

At the start of the project, no additional hardware or software purchases are required. Nahayat provides the hardware and test environment required for the initial research, development and benchmarking activities.

Comparative testing with commercial cloud-based LLMs may result in usage-based API costs through the company's existing services. These costs will be considered where relevant during the financial comparison.

Potential investment in additional hardware is not considered an initial project expense. Instead, the required hardware and associated costs will be investigated as part of the Hardware Requirements Analysis and Financial Feasibility Analysis.

Based on the technical benchmark results and expected usage, the project will evaluate whether such an investment can be financially justified. Any actual hardware purchase falls outside the scope of this project unless separately approved by the company.

4.2 Risks and fall-back activities

Risk	Prevention activities included in plan	Fall-back Activities
1 Available hardware is insufficient for certain models or use cases.	insufficient for certain models or use cases. Research model requirements and test different model sizes and quantization levels while monitoring hardware utilization.	Test smaller/quantized models. If still infeasible, document the limitation and determine what hardware would be required.
2 The scope becomes too broad due to the four use-case categories.	Use clearly defined benchmark scenarios and limit the number of models tested in depth.	Prioritize experiments based on relevance. Any reduction of the use cases will be discussed with the company supervisor and supported by technical evidence.
3 Local models provide insufficient quality or reliability.	Evaluate multiple suitable models using consistent test cases and criteria.	Document the limitation and recommend cloud-based or hybrid inference where local models are unsuitable.

5. Other

The following tools are used to support the planning, documentation, version control, and progress tracking of the project:

Jira Board: [Jira Link](#)

Used for project planning, task management, sprint tracking, and monitoring project progress.

GitHub Repository: [Github Repo Link](#)

Used for version control and for storing project documentation, benchmark scripts, configurations, test results, and prototype implementations.

5.1 Glossary and Acronyms

Term / Acronym	Meaning
Ai	Artificial Intelligence – Computer systems designed to perform tasks that normally require human intelligence.
API	Application Programming Interface – A way for different software systems to communicate with each other. In this project, APIs are used to access externally hosted LLMs and other services.
MCP	Model Context Protocol – A protocol that allows AI applications and models to interact with external tools, systems and data sources through a standardized interface.
RAG	Retrieval-Augmented Generation – An approach in which relevant information is retrieved from an external knowledge source and provided to an LLM as additional context before generating a response.
SMB	Small and Medium-sized Business – A business that falls within the small or medium-sized enterprise category.
Agentic Tool Use	The ability of an AI system or LLM to select and use external tools or functions to complete a task.
Tool Calling	The capability of an LLM to request the execution of an external function or tool as part of completing a task.